Programming Principles 1 (PP1) — Progress Tracker & Roadmap

Second-Phase Progression: Arrays → Strings, Attestation Checkpoint, and Quiz 3 Scope

Fall 2026 | Lecturer: **Nikita**

W5: 1D Arrays [OK] ▸ W6: 2D Arrays [OK] ▸ W7: Quiz 2 [OK] ▸ **W8: STRINGS [HERE]** ▸ W9: Functions ▸ W10: STL 1 ▸ W11: Quiz 3 (20p)

WEEK 05 — COMPLETED

L4: 1D Arrays & Streams

- Contiguous memory & 0-indexing
- Linear search, min/max patterns
- sort(a, a+n) and reverse
- freopen: comment it before Ejudge

Evaluated in Quiz 2

WEEK 06 — COMPLETED

L5: 2D Arrays (Matrices)

- Nested loops & the index grid
- Diagonals, symmetry, patterns
- Sorting tiers: rows → full → pointer
- O(n²) complexity, measured live

Evaluated in Quiz 2

WEEK 07 — COMPLETED

Milestone: Quiz 2 (15 pts)

- Contest held: Sunday, Oct 18
- Scope: cumulative L1–L5
- Labs 3, 4, 5 closed on Oct 17
- Score feeds Attestation 1

Result Recorded

CHECKPOINT — ATTESTATION 1

Attestation 1 · 25 pts

- **Covers:** Labs 1–5 + Quiz 1 + Quiz 2
- WSP attestation: check Monday 26.10
- Labs 1–5 are closed — no late work
- Lab 6 is open: due before next lecture

Att1 25 + Att2 45 = 70 → cap 60

WEEK 08 — ACTIVE NOW

L6: Strings

- Construct, size() / length()
- Copy, compare, concatenate
- Indexing & range-for, `char &` to edit
- char ↔ number, case conversion
- find, substr, erase/insert, stringstream

Target: Lab 6 Submission

WEEK 09 — UPCOMING

L7: Functions

- Declare, define, call
- Pass by value vs. reference
- 1D and 2D arrays as parameters
- Local vs. global scope

Target: Lab 7

WEEK 10 — UPCOMING

L8: STL Part 1

- vector, set, map
- Iterators — finally defined
- sort with a custom comparator
- Frequency counting with map

Target: Lab 8

WEEK 11 — MAJOR EXAM

Milestone: Quiz 3 Contest

- **Scope:** cumulative L1–L8
- **Weight:** 20 pts — 4 tasks × 5
- **Date:** Sunday, Nov 15
- **Labs 6, 7, 8:** due 14.11 at 23:59

Cumulative Attestation

SCORING UPDATE **Quiz 3 & Quiz 4 — 20 pts each** (4 tasks × 5, +5 bonus) · **Att2 max: 45 pts** · **semester cap stays 60** — points above 60 do not carry over to the exam.

STUDENT ACTION DIRECTIVE FOR WEEK 08

Do not leave Lab 6 for the quiz week

- **Focus:** Lab 6 — strings, not arrays.
- **Reading:** Ch. 6 (Strings) in the Manual.
- **Drill:** palindrome, case shift, split a line.
- **Habit:** `for (char &c : s)` when you edit in place.
- **Rule:** never submit with `freopen` uncommented.
- **Next:** Week 09 turns these loops into functions.